FINAL ASSESSMENT TEST

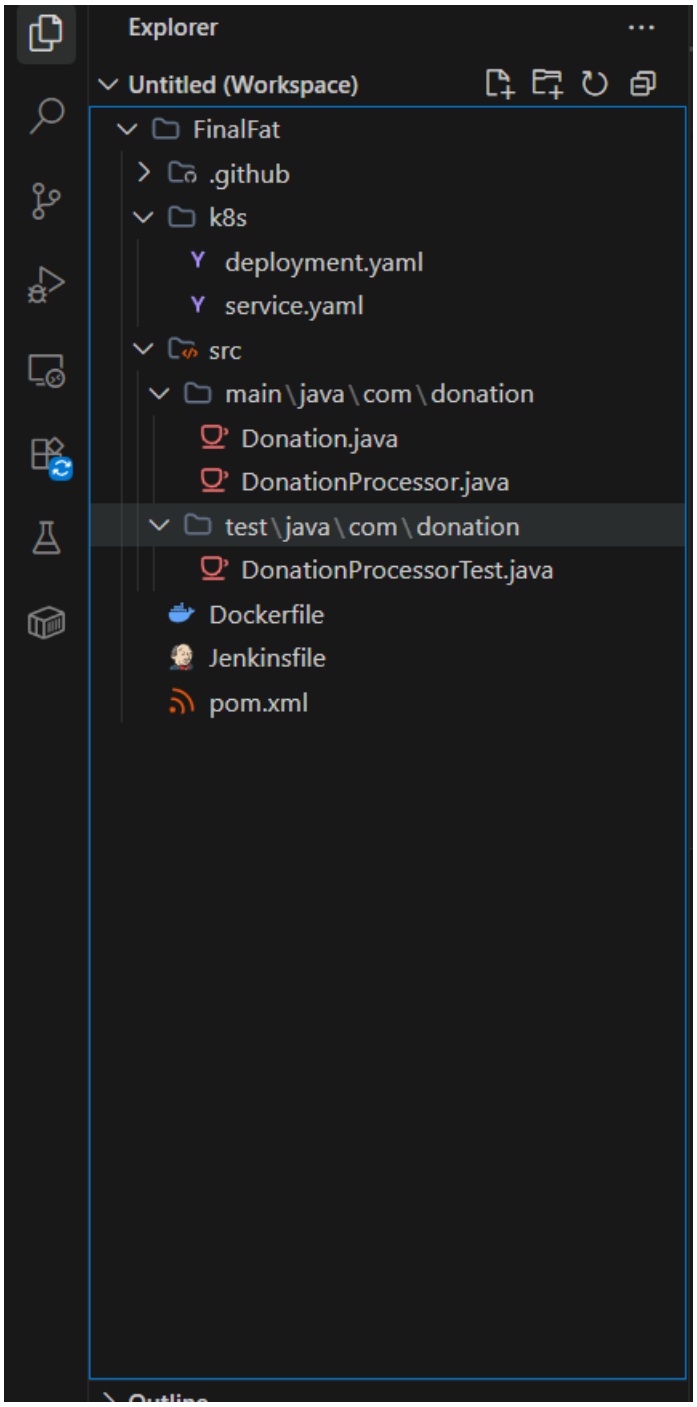
AGILE LAB

NAME: TAMIL SELVAN P

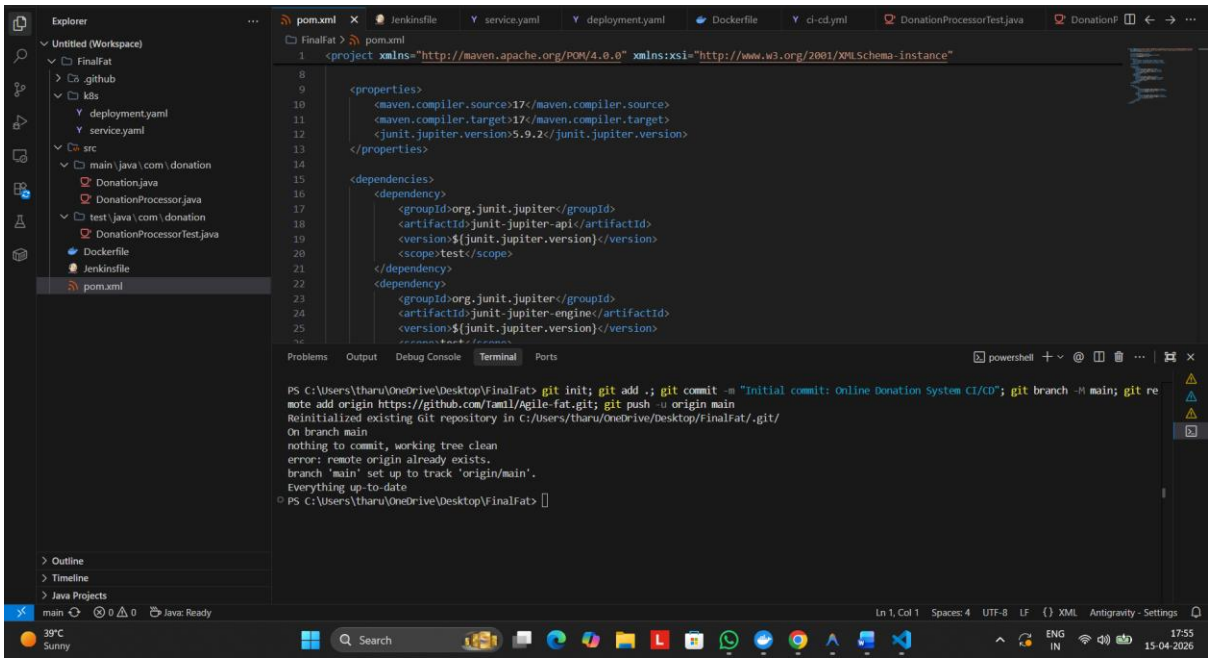
REGNO: 23MIS0214

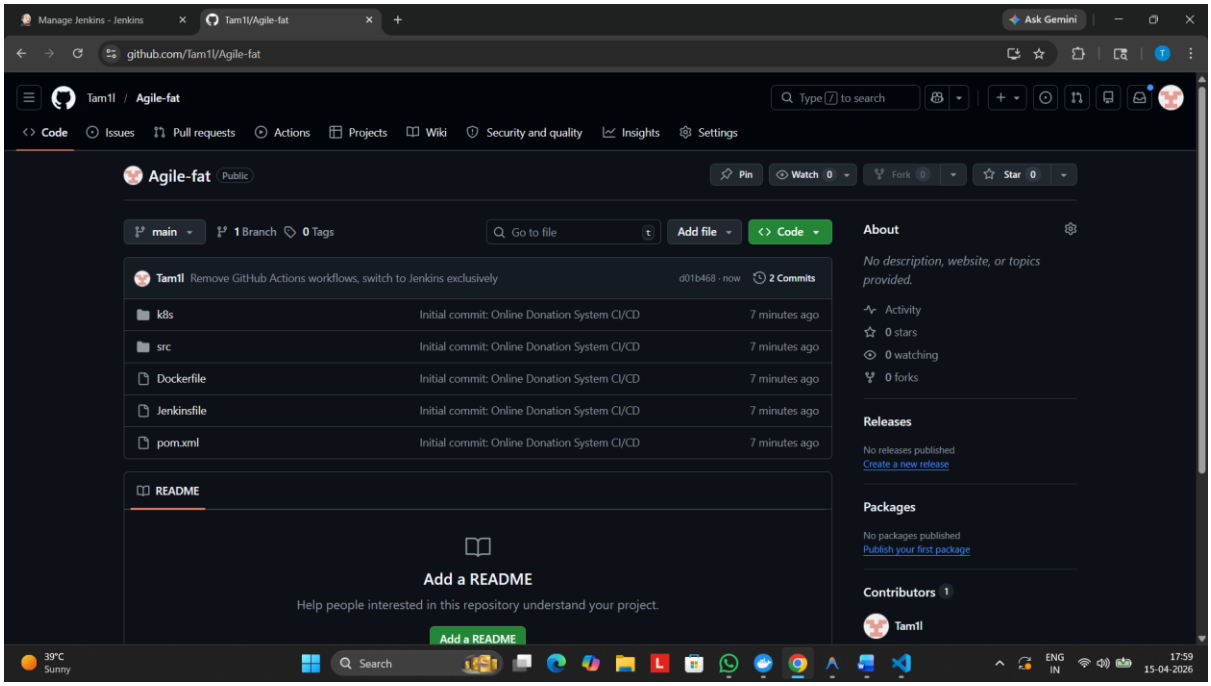
SET: DONATION ARE PROCESSED AND RECORDED ACCURATELY

File Structure:

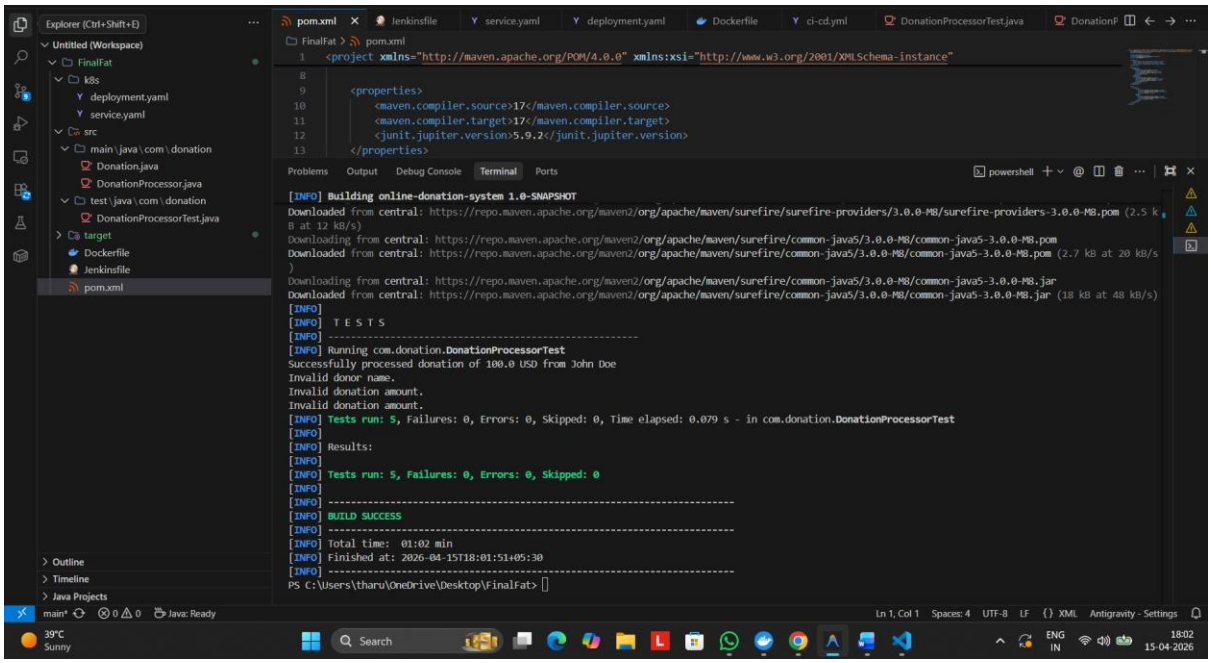


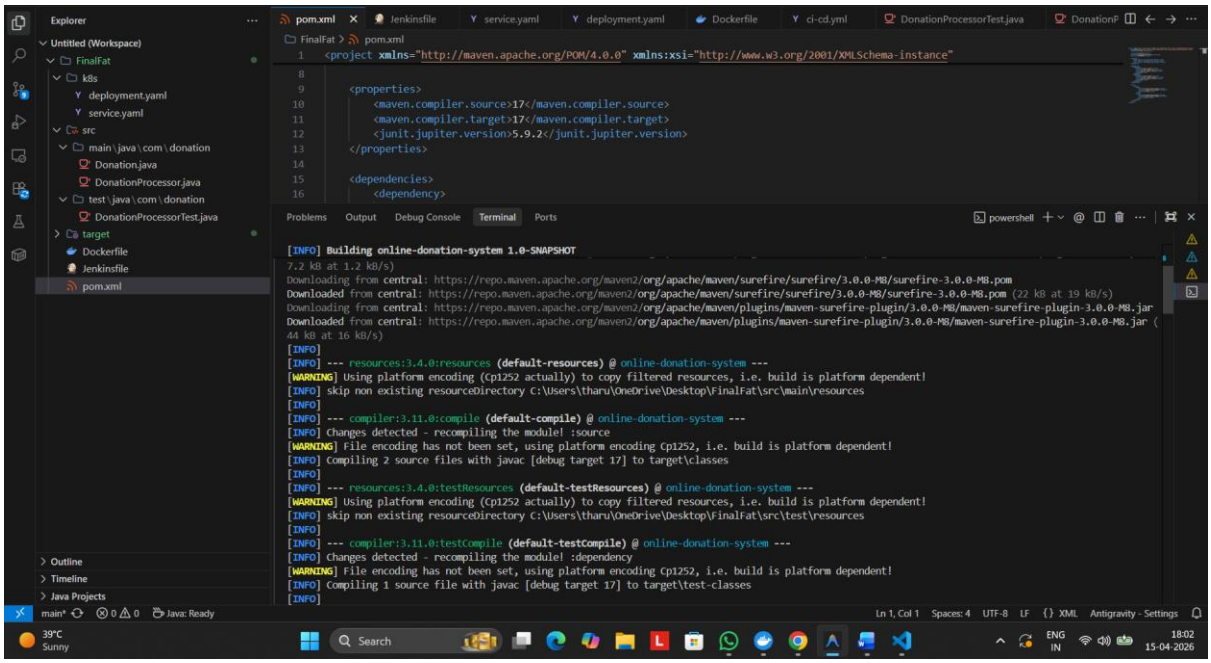
Pushing the Code to the Github





Junit test:


pom.xml
Jenkinsfile
service.yaml
deployment.yaml
Dockerfile
ci-cd.yml
DonationProcessorTest.java
FinalFat > pom.xml
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
9 <properties>
10 <maven.compiler.source>17</maven.compiler.source>
11 <maven.compiler.target>17</maven.compiler.target>
12 <junit.jupiter.version>5.9.2</junit.jupiter.version>
13 </properties>
Problems Output Debug Console Terminal Ports
[INFO] Building online-donation-system 1.0-SNAPSHOT
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.0.0-M8/surefire-providers-3.0.0-M8.pom (2.5 kB at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.0.0-M8/common-java5-3.0.0-M8.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.0.0-M8/common-java5-3.0.0-M8.pom (2.7 kB at 20 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.0.0-M8/common-java5-3.0.0-M8.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.0.0-M8/common-java5-3.0.0-M8.jar (18 kB at 48 kB/s)
[INFO] T E S T S
[INFO] ---
[INFO] Running com.donation.DonationProcessorTest
Successfully processed donation of 100.0 USD from John Doe
Invalid donor name.
Invalid donation amount.
Invalid donation amount.
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.079 s - in com.donation.DonationProcessorTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 01:02 min
[INFO] Finished at: 2026-04-15T18:01:51+05:30
PS C:\Users\tharu\OneDrive\Desktop\FinalFat>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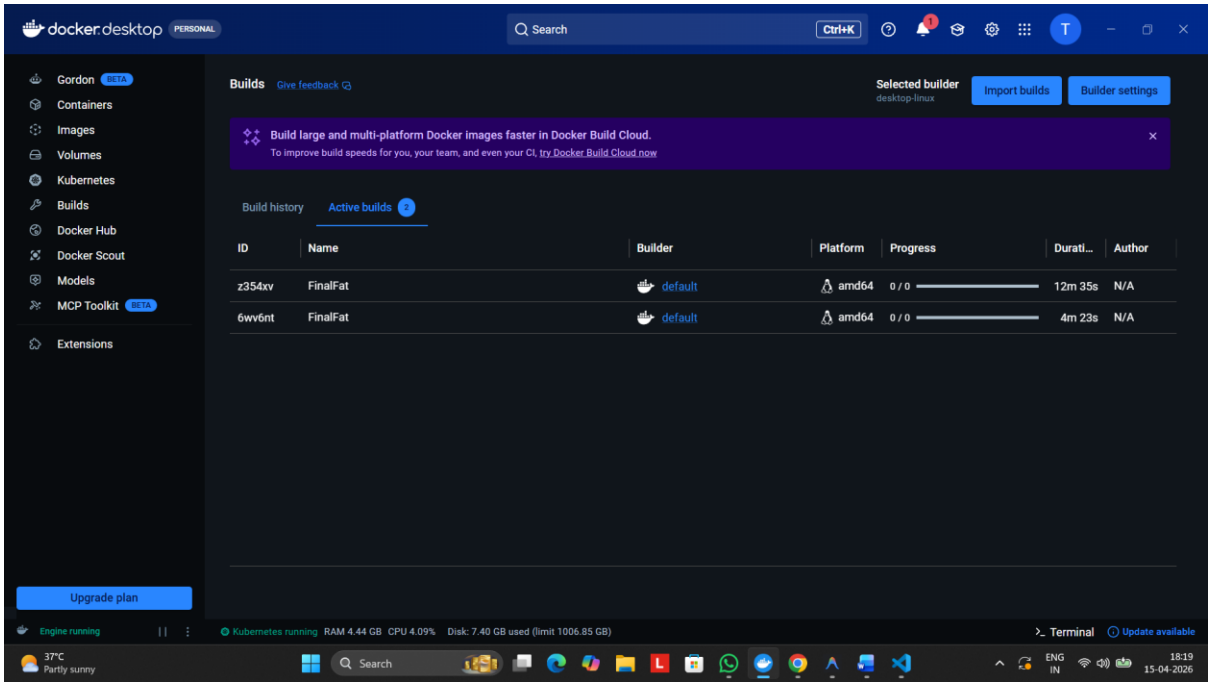
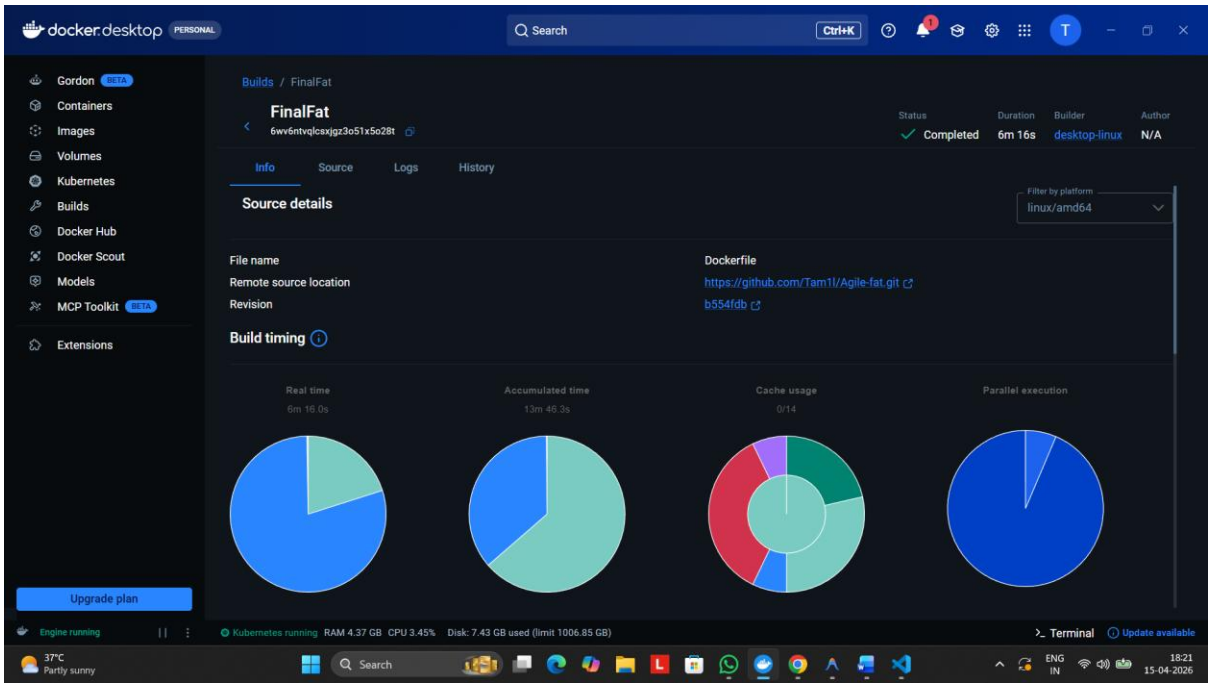
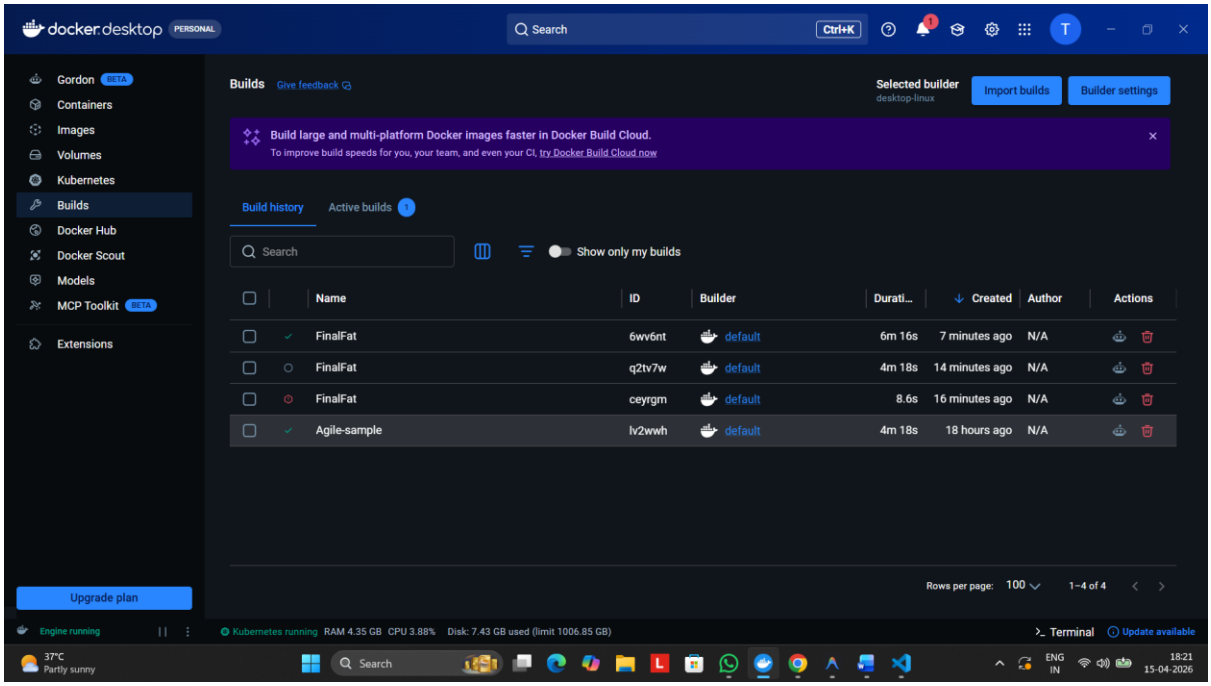
12 <junit.jupiter.version>5.9.2</junit.jupiter.version>
13 </properties>
15 <dependencies>
16 <dependency>
Problems Output Debug Console Terminal Ports
[INFO] Building online-donation-system 1.0-SNAPSHOT
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.0.0-M8/surefire-3.0.0-M8.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.0.0-M8/surefire-3.0.0-M8.pom (22 kB at 19 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.0.0-M8/maven-surefire-plugin-3.0.0-M8.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.0.0-M8/maven-surefire-plugin-3.0.0-M8.jar (44 kB at 16 kB/s)
[INFO] --- resources:3.4.0:resources (default-resources) @ online-donation-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\tharu\OneDrive\Desktop\FinalFat\src\main\resources
[INFO] --- compiler:3.11.0:compile (default-compile) @ online-donation-system ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 2 source files with javac [debug target 17] to target\classes
[INFO] --- resources:3.4.0:testResources (default-testResources) @ online-donation-system ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\tharu\OneDrive\Desktop\FinalFat\src\test\resources
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ online-donation-system ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes

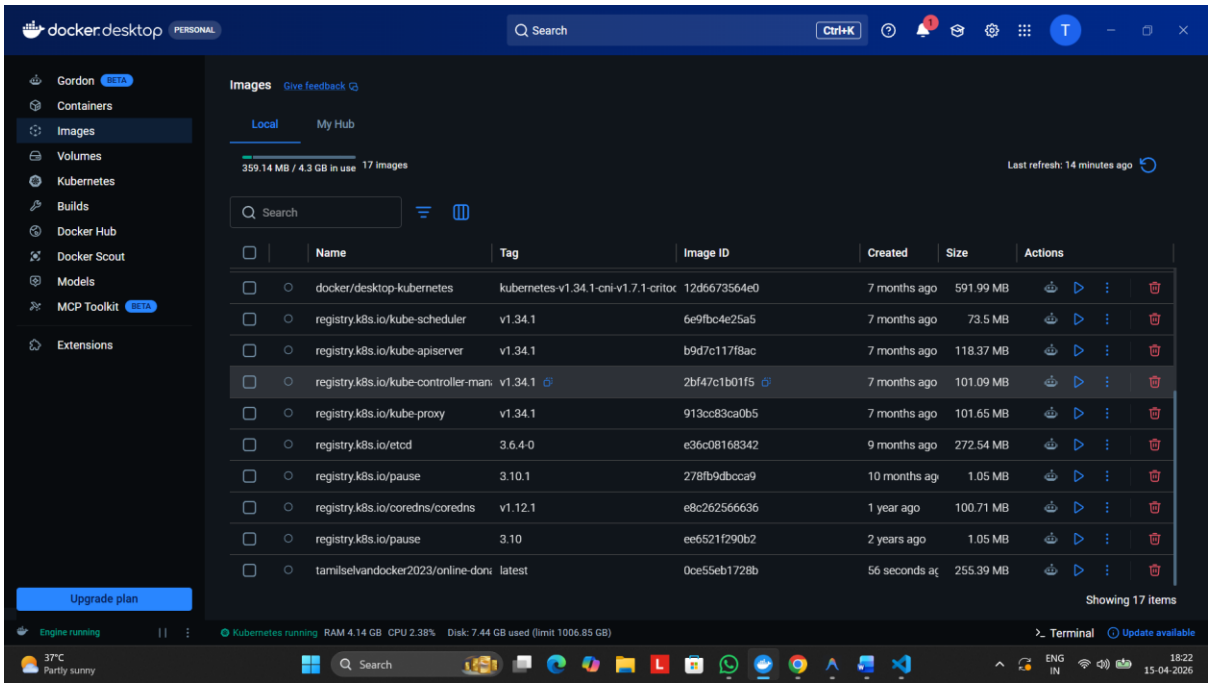
Docker After the build (Container and the push):

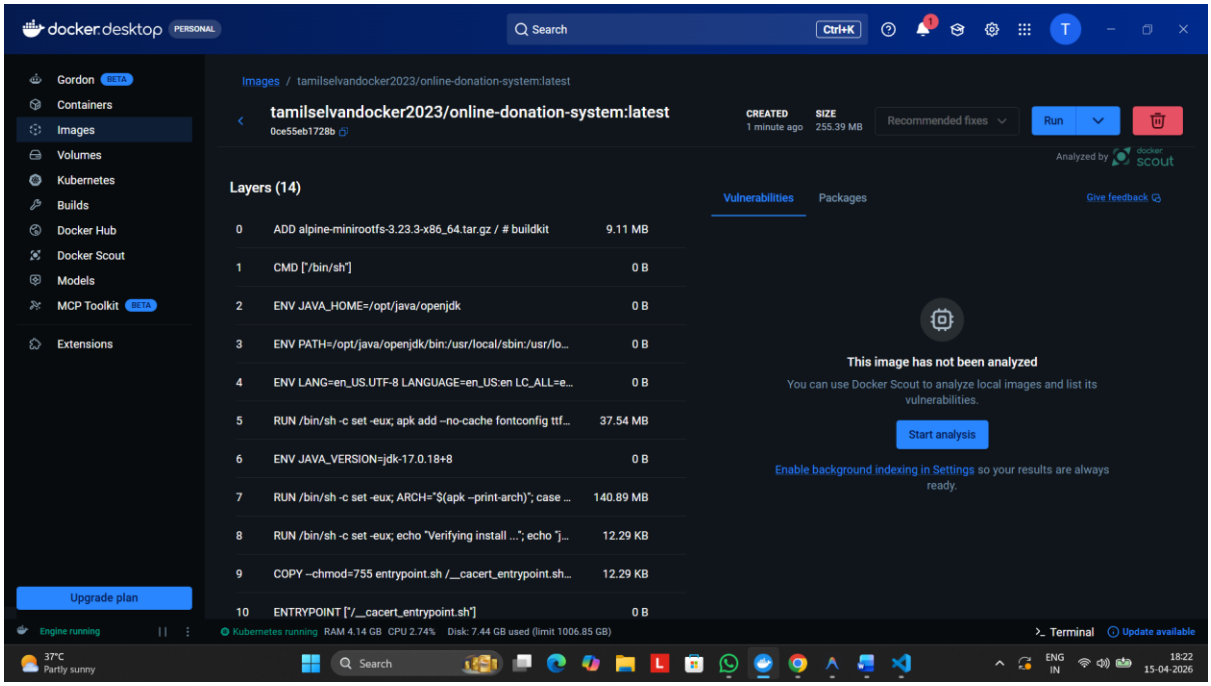




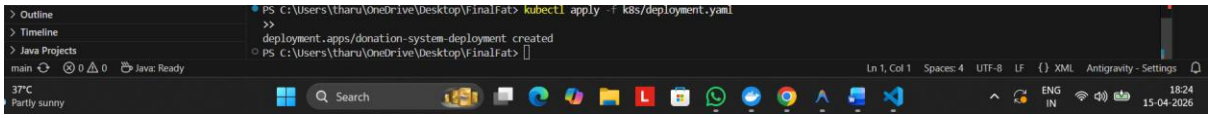


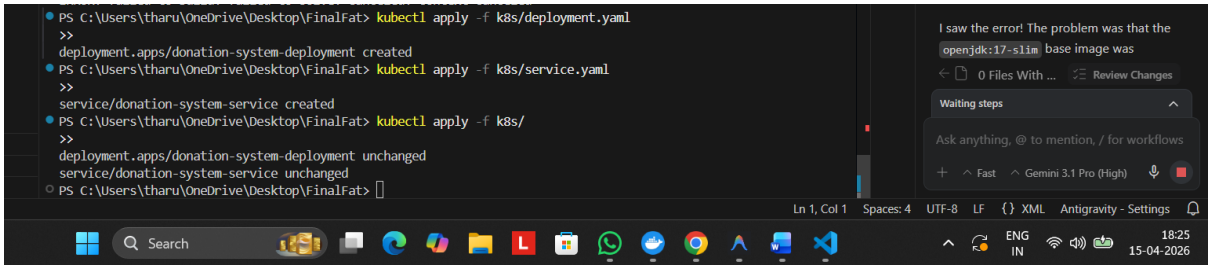
Docker Image pushing:

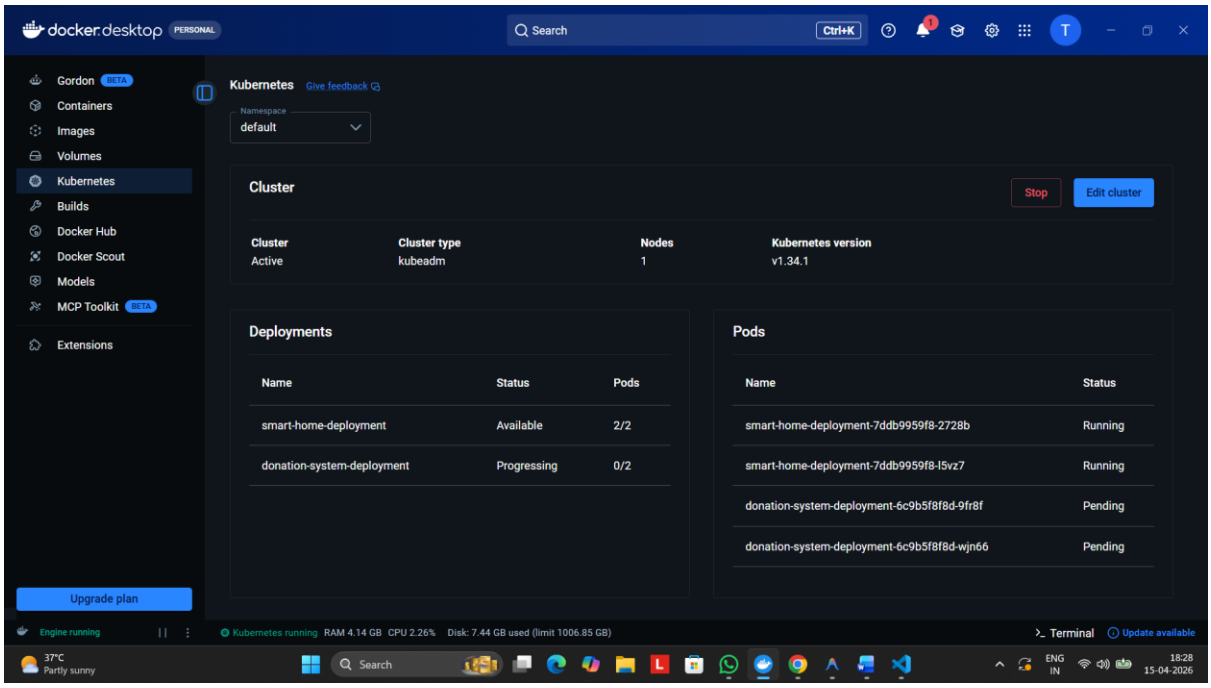




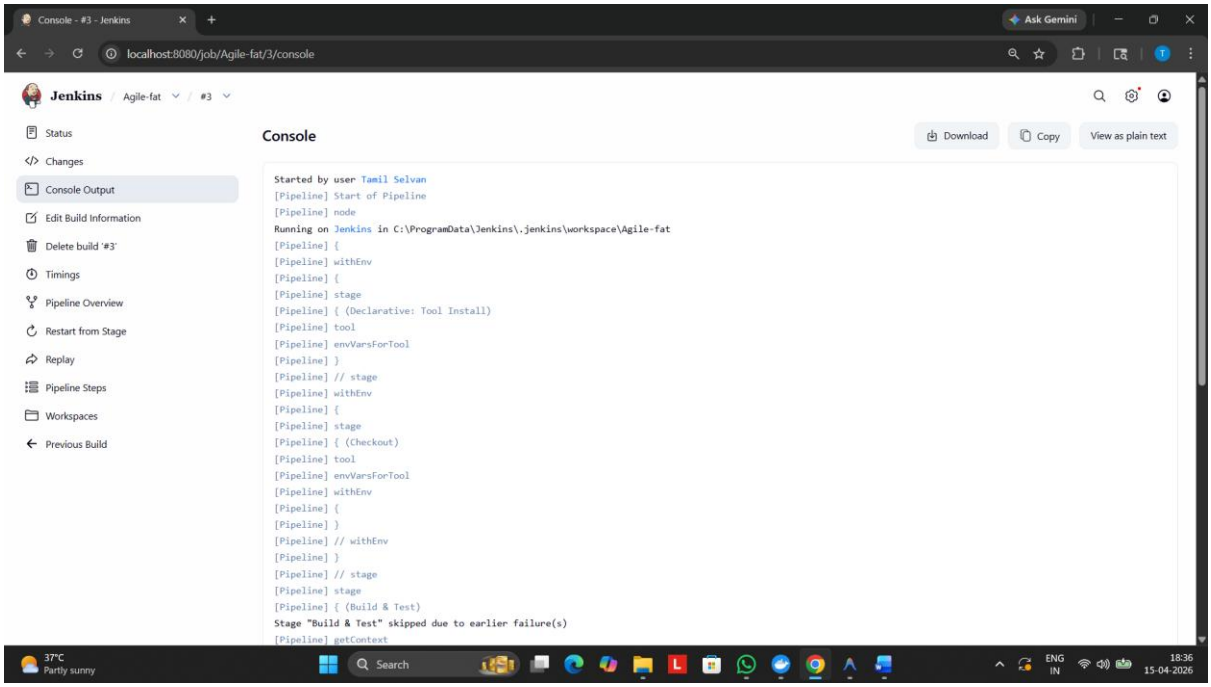
Kubernetes Deployment:

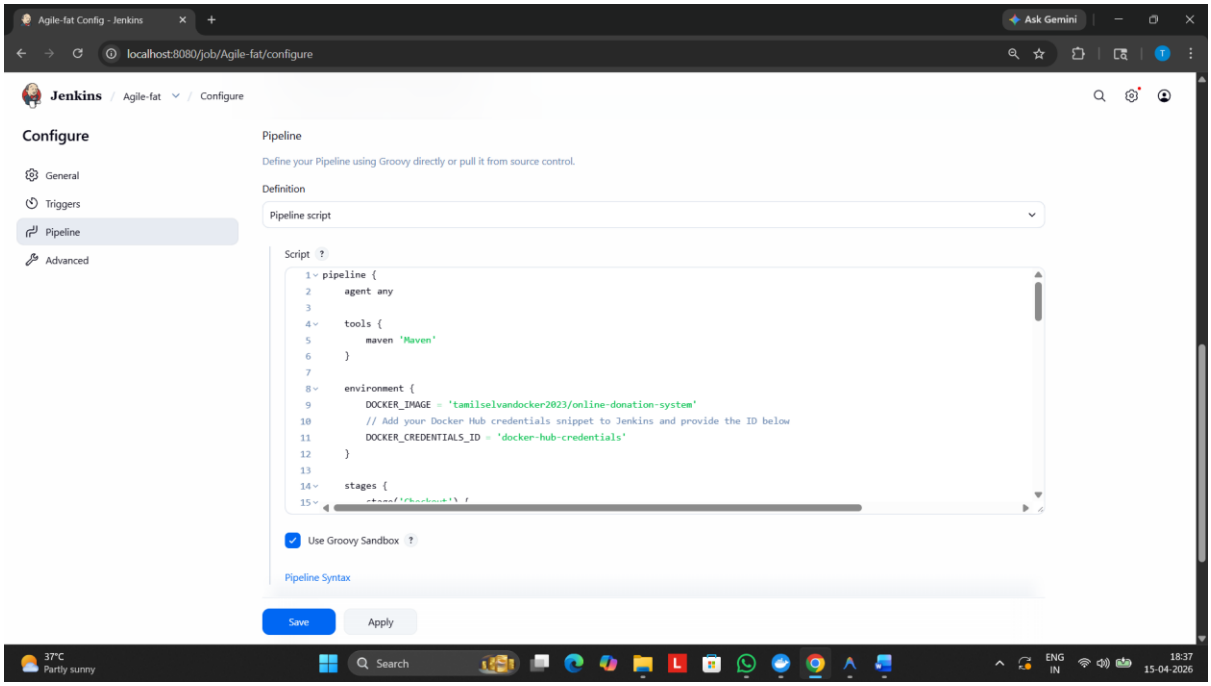






Jenkins CI/CD Pipeline:



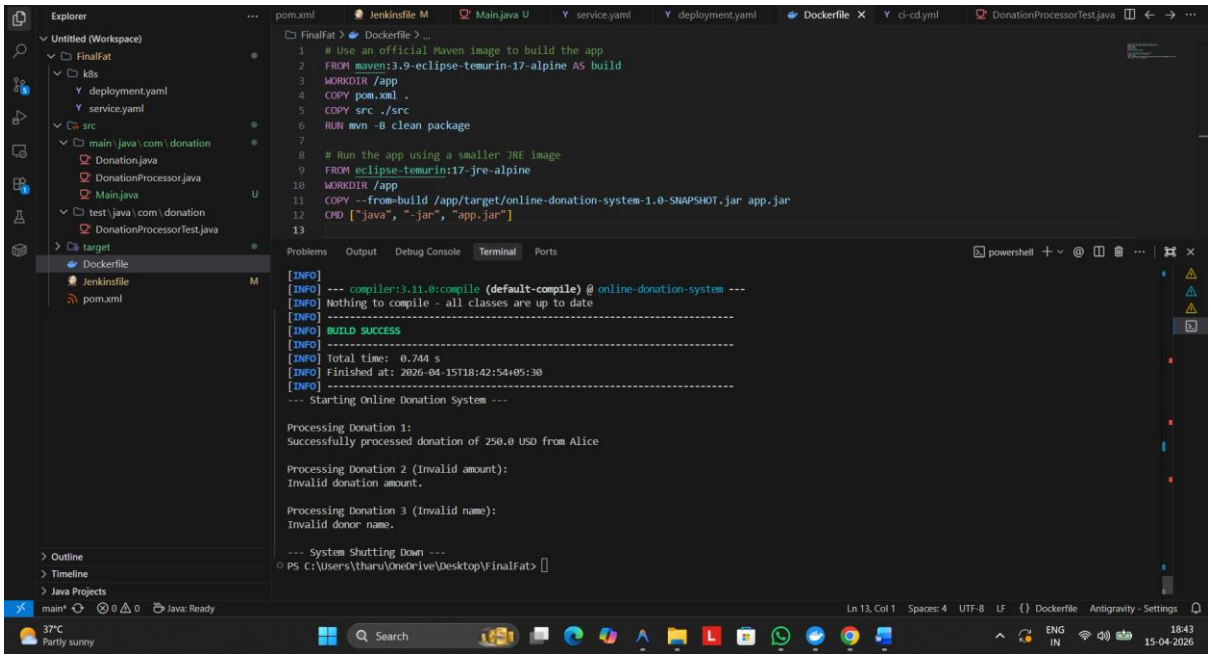


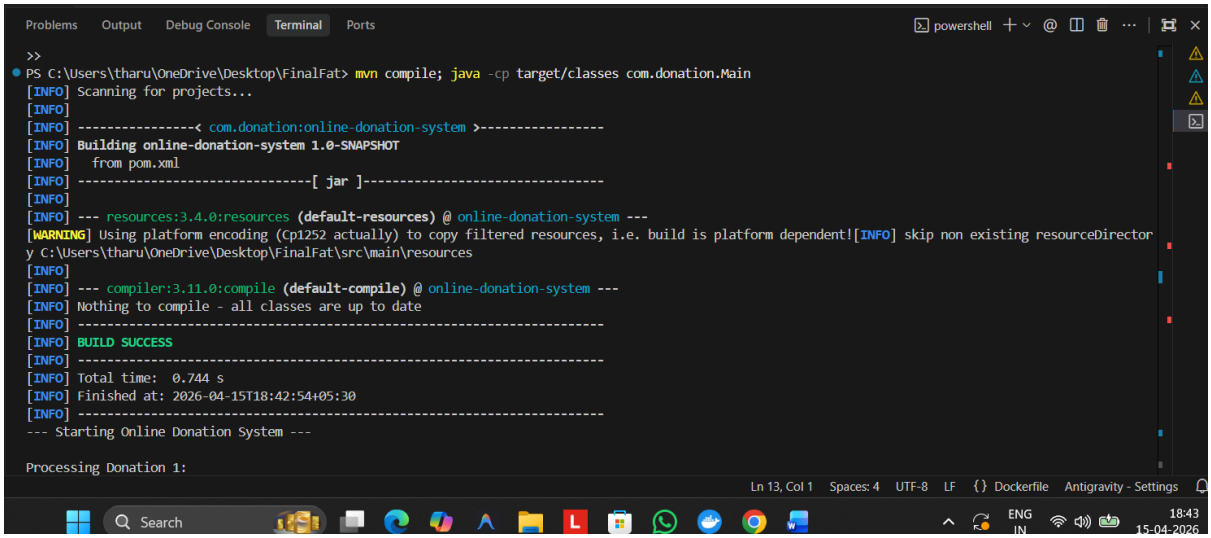

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Docker Run)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\auction-system>docker run ticket-system
The highest bidder is: Bidder: Bob, Amount: 150.0
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

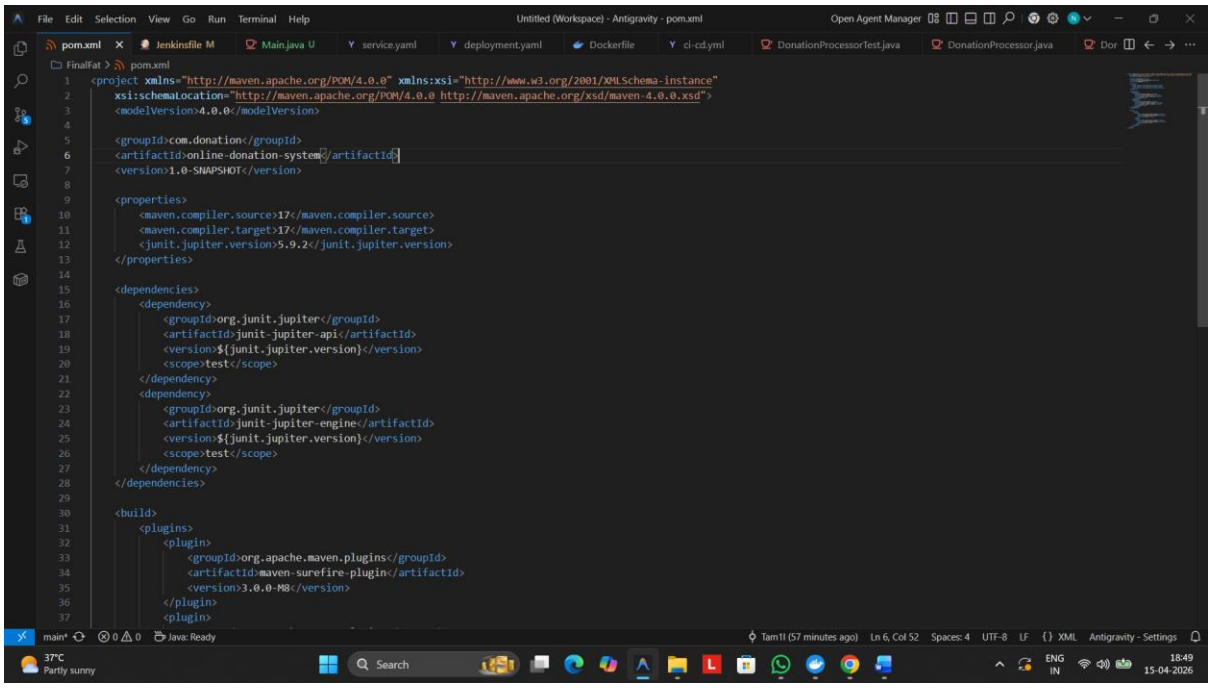
Jenkins 2.541.3

Java Program Output:





Code:



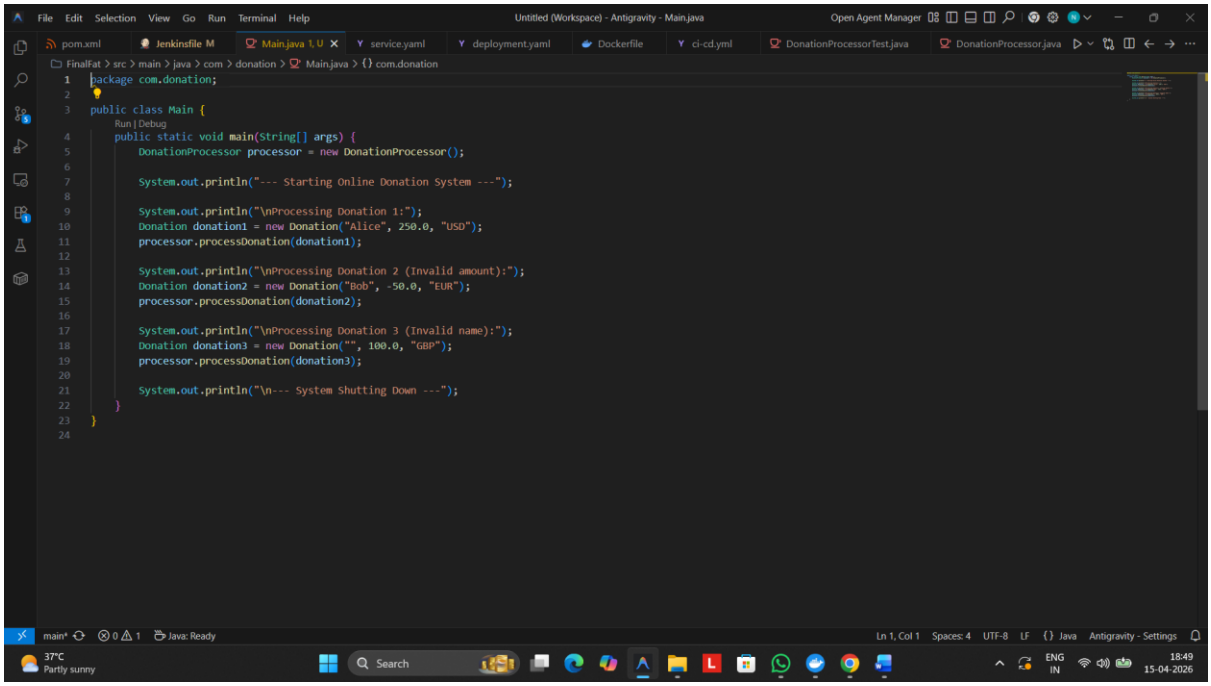
```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.donation</groupId>
    <artifactId>online-donation-system</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <junit.jupiter.version>5.9.2</junit.jupiter.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-api</artifactId>
            <version>${junit.jupiter.version}</version>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-engine</artifactId>
            <version>${junit.jupiter.version}</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0-M8</version>
            </plugin>
            <plugin>
```



```java
package com.donation;

public class Main {
    public static void main(String[] args) {
        DonationProcessor processor = new DonationProcessor();

        System.out.println("--- Starting Online Donation System ---");

        System.out.println("\nProcessing Donation 1:");
        Donation donation1 = new Donation("Alice", 250.0, "USD");
        processor.processDonation(donation1);

        System.out.println("\nProcessing Donation 2 (Invalid amount):");
        Donation donation2 = new Donation("Bob", -50.0, "EUR");
        processor.processDonation(donation2);

        System.out.println("\nProcessing Donation 3 (Invalid name):");
        Donation donation3 = new Donation("", 100.0, "GBP");
        processor.processDonation(donation3);

        System.out.println("\n--- System Shutting Down ---");
    }
}
```

File Edit Selection View Go Run Terminal Help — Untitled (Workspace) - Antigravity - service.yaml

FinalFat > k8s > service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: donation-system-service
spec:
  type: LoadBalancer
  selector:
    app: donation-system
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
```

File Edit Selection View Go Run Terminal Help — Untitled (Workspace) - Antigravity - deployment.yaml

FinalFat > k8s > deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: donation-system-deployment
  labels:
    app: donation-system
spec:
  replicas: 2
  selector:
    matchLabels:
      app: donation-system
  template:
    metadata:
      labels:
        app: donation-system
    spec:
      containers:
      - name: donation-system
        image: tamilselvandocker2023/online-donation-system:latest
        ports:
        - containerPort: 8080
```

```dockerfile
# Use an official Maven image to build the app
FROM maven:3.9-eclipse-temurin-17-alpine AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn -B clean package

# Run the app using a smaller JRE image
FROM eclipse-temurin:17-jre-alpine
WORKDIR /app
COPY --from=build /app/target/online-donation-system-1.0-SNAPSHOT.jar app.jar
CMD ["java", "-jar", "app.jar"]
```

```java
package com.donation;

import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class DonationProcessorTest {

    private DonationProcessor processor;

    @BeforeEach
    public void setUp() {
        processor = new DonationProcessor();
    }

    @Test
    public void testValidDonation() {
        Donation donation = new Donation("John Doe", 100.0, "USD");
        assertTrue(processor.processDonation(donation), "Valid donation should be processed successfully");
    }

    @Test
    public void testZeroAmountDonation() {
        Donation donation = new Donation("Jane Doe", 0.0, "USD");
        assertFalse(processor.processDonation(donation), "Zero amount donation should fail");
    }

    @Test
    public void testNegativeAmountDonation() {
        Donation donation = new Donation("Bob", -50.0, "EUR");
        assertFalse(processor.processDonation(donation), "Negative amount donation should fail");
    }

    @Test
    public void testEmptyDonorName() {
        Donation donation = new Donation("", 100.0, "USD");
        assertFalse(processor.processDonation(donation), "Empty donor name should fail");
```

```java
package com.donation;

public class DonationProcessor {

    public boolean processDonation(Donation donation) {
        if (donation == null) {
            throw new IllegalArgumentException("Donation cannot be null");
        }

        if (donation.getAmount() <= 0) {
            System.out.println("Invalid donation amount.");
            return false;
        }

        if (donation.getDonorName() == null || donation.getDonorName().trim().isEmpty()) {
            System.out.println("Invalid donor name.");
            return false;
        }

        // Simulate successful processing
        System.out.println("Successfully processed donation of " + donation.getAmount() + " " + donation.getCurrency() + " from " + donation.getDonorName());
        return true;
    }
}
```

```java
package com.donation;

public class Donation {
    private String donorName;
    private double amount;
    private String currency;

    public Donation(String donorName, double amount, String currency) {
        this.donorName = donorName;
        this.amount = amount;
        this.currency = currency;
    }

    public String getDonorName() { return donorName; }
    public void setDonorName(String donorName) { this.donorName = donorName; }

    public double getAmount() { return amount; }
    public void setAmount(double amount) { this.amount = amount; }

    public String getCurrency() { return currency; }
    public void setCurrency(String currency) { this.currency = currency; }
}
```